AHMAD IRFAN FARDANI

ELECTRONICS ENGINEERING

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Portfolio: https://irfan33223102.github.io/porotofolio/
Semarang, Central Jawa, Indonesia

D3 Electronics Engineering graduate from Politeknik Negeri Semarang with a deep interest in Embedded Systems, IoT development, and hardware design. Experienced in microcontroller programming, PCB design, and industrial automation through academic projects and industry internships. Capable of developing efficient, reliable, and practical electronics solutions. Possesses strong analytical skills in technical problem-solving and systematic electronics troubleshooting.

EDUCATION

Politeknik Negeri Semarang **Semarang, Indonesia**
Diploma III in Electronics Engineering | GPA: 3.49 / 4.00 | Accreditation: Excellent *2023 – 2026*
Final Project: IoT-Based River Monitoring System for Early Flood Warning (Rainfall, Water Level & Sediment)
Competencies: Microcontroller, Industrial Automation, Control Systems

SMA Negeri 11 Semarang **Semarang, Indonesia**
Natural Sciences (IPA) | Major: Physics and Advanced Mathematics *2020 – 2023*

PROJECT EXPERIENCE

IoT-Based River Monitoring System (Rainfall, Water Level & Sediment) **2026**
Final Project – Politeknik Negeri Semarang

- Designed a real-time IoT monitoring system for three critical river parameters (rainfall, water level, sediment) for early flood detection.
- Utilized ESP32-WROOM-32; data transmitted to cloud server via MQTT protocol and displayed on SIMBAS web dashboard.
- Integrated AJ-SR04M ultrasonic sensor, Rain Gauge (Tipping Bucket), Load Cell + HX711, and automated sediment sampling mechanism (Peristaltic Pump, Servo Motors).

MONIK – Web-Based Hydroponic Monitoring System **2025**
Internship Project – CV. Omah IoT

- Developed integrated monitoring for pakcoy hydroponic cultivation, tracking pH, EC, water temperature, humidity, and light intensity in real-time.
- Built with Arduino Uno R3 WiFi and ESP8266; displayed locally on LCD 20x4 and remotely via web server dashboard.

IoT-Based Aquarium Temperature Monitoring & Control **2024**
Academic Project – Politeknik Negeri Semarang

- Designed automatic aquarium temperature regulation using NodeMCU ESP8266 with Telegram Bot for real-time remote monitoring and control.
- System activates water pump when temperature is too high and heater when too low; users can set setpoint via Telegram Bot.

PID-Based Water Level Control System for Fish Farm Ponds **2026**
Academic Project – Politeknik Negeri Semarang

- Designed automatic water level control system using Arduino Uno with PID (Proportional-Integral-Derivative) algorithm.
- Integrated HY-SRF05 ultrasonic sensor for level measurement and DC pump control to maintain stable setpoint.

Multifunctional Variable DC Power Supply **2025**

Internship Project – CV. Omah IoT

- Designed and assembled a variable power supply (1.2V–24V) with short circuit detector and digital volt-ampere meter.
- Strengthened skills in schematic reading, wiring assembly, and power electronics troubleshooting.

INTERNSHIP EXPERIENCE

CV. Omah IoT **Semarang, Indonesia**

Hardware Development – IoT | Industry Internship *September 2025 – January 2026*

- Programmed microcontrollers using Arduino IDE for various industrial IoT projects.
- Assembled hardware enclosures and industrial control panel/box units.
- Designed circuit schematics and PCB layouts using Proteus, EasyEDA, and Eagle.
- Performed troubleshooting and optimization of electronics systems.
- Collaborated with team in end-to-end IoT product development.

Result: Excellent (Sangat Baik)

CERTIFICATION

Competency Certificate – Industrial Automation Technician **17 September 2025**

Badan Nasional Sertifikasi Profesi (BNSP) | No. 78411.3119 0.0006999 2025 | Valid: 2025 – 2028

Industry Internship Certificate **Agustus 2025 - Januari 2026**

CV. Omah IoT | No. 089/0109.3101/ITR/XI/2026 | Position: Hardware Development – IoT

LANGUAGE

English

Bahasa Indonesia

SKILLS

Hard Skills Microcontroller Programming (Arduino IDE), PCB Design (Proteus, EasyEDA, Eagle), IoT System Integration (Sensor & Actuator), Industrial Automation, PID Control Systems, Electrical & Electronics Component Measurement, MQTT Protocol, Web Server (ESP8266/ESP32).

Soft Skills Teamwork, Problem-Solving, Analytical Thinking, Attention to Detail, Fast Adaptation to New Technologies.

Software Arduino IDE, Proteus, EasyEDA, Eagle, MATLAB, CX – Progamer, CX – Desaigner, Visual Studio Code, Ms. Word, Ms. Excel, Canva.